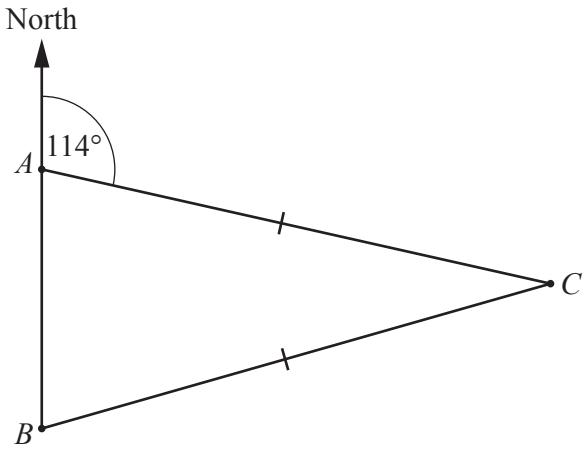
0580_s23_qp_43_Q4

4 **(a)**



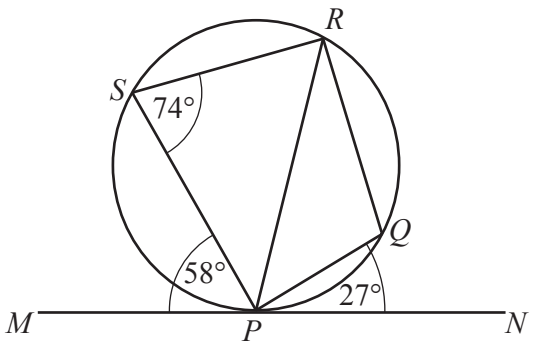
NOT TO SCALE

A, B and C are three towns and the bearing of C from A is 114°.
B is due south of A and $AC = BC$.

Calculate the bearing of B from C.

..... [3]

(b)



NOT TO SCALE

P, Q, R and S lie on a circle.
MPN is a tangent to the circle at P.
Angle $MPS = 58°$, angle $PSR = 74°$ and angle $QPN = 27°$.

(i) Find angle PRS.

Angle PRS = ... [1]

(ii) Find angle PQR.

Angle PQR = ... [1]

(iii) Find angle RPQ.

Angle RPQ = ... [2]

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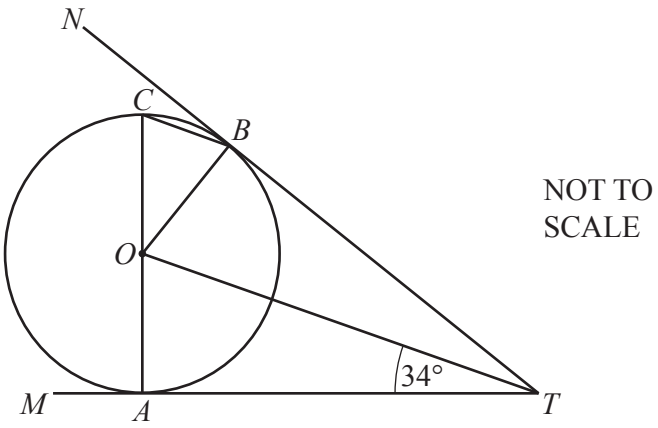
(c)



A, *B* and *C* lie on a circle, centre *O*, with diameter *AC*.
TAM and *TBN* are tangents to the circle and angle *ATO* = 34°.

Using values and geometrical reasons, complete these statements to show that *CB* is parallel to *OT*.

In triangles *AOT* and *BOT*, *OT* is common.
Angle *OAT* = angle *OBT* = 90° because .....

.....

AT = *BT* because .....

.....

Triangle *AOT* is congruent to triangle *BOT* because of congruence criterion .....

Angle *AOT* = angle *BOT* = 56° because angles in a triangle add up to 180°.

Angle *BOC* = ... ° because .....

Angle *OBC* = ... ° because .....

.....

CB is parallel to *OT* because .....

[6]